Java OOP In-Class Test - Questions and Answers

Object Oriented Programming (Java) · October 2024 · typed up from test review screenshots

Screenshots taken 17 October 2024 from another student's Blackboard review page, so this is not one of the Level 4 modules in this repository (most likely a Level 5 OOP module). Question numbers are as shown; Q4 and Q14 were not in the screenshots. The answer key at the end was worked out and then confirmed by compiling and running the code; several answers selected in the screenshots were wrong.

Part A - Multiple choice

Question 1

```
class T1 extends Thread {
    public void run() {        // line 2
        while (true)
            System.out.println("Thread T1");
    }
}

class T2 extends Thread {
    public void run() {        // line 10
        while (true)
            System.out.println("Thread T2");
    }
}

class ThreadsQ15 {
    public static void main(String[] args) {
        T1 t1 = new T1();      // line 18
        T2 t2 = new T2();      // line 19
        t1.run();              // line 20
        t2.start();            // line 21
    }
}
```

The following code was written by a developer in order to display for ever interleaved (mixed) messages of both Thread T1 and Thread T2. However, the code just displays the message Thread T1 for ever. Which of the following changes should be made to fix the code to run as intended?

- **A.** Swap the order of lines 19 and 20
- **B.** Replace both lines 2 and 10 with: public void start()
- **C.** Replace line 10 with: public void start()
- **D.** Replace line 20 with: t1.start();
- **E.** Replace line 21 with: t2.run();

Question 2

Which of the following is NOT a characteristic of object oriented programming?

- **A.** Instructional decomposition
- **B.** Polymorphism
- **C.** Abstraction
- **D.** Encapsulation

Question 5

```
class Q7 {
    int x;
```

```
    Q7(int x1) {
        x = x1;
    }

    public static void main(String[] arg) {
        Q7 q1 = new Q7(100);
        Q7 q2 = new Q7(100);

        if (q1.equals(q2))
            System.out.println(true);
        else
            System.out.println(false);

        String str1 = new String("123");
        String str2 = new String("123");

        if (str1.equals(str2))
            System.out.println(true);
        else
            System.out.println(false);
    }
}
```

What is the output of the following program?

A. false
false

B. false
true

C. The program will throw an exception during its execution.

D. true
true

E. true
false

Question 6

```
interface F {
    void foo();
}

class Y implements F {
    public void foo() {
        System.out.print("Y ");
    }
}

public class Q6 extends Y {
    public void foo() {
        System.out.print("Q6 ");
    }

    public static void main(String[] args) {
        F f1 = new Q6();
        f1.foo();
        F f2 = new Y();
        f2.foo();
        Y y1 = new Q6();
        y1.foo();
    }
```

```
}
```

What is the output of the following program?

A. Q6 Y Y

B. Q6 Y Q6

C. Y Y Q6

D. Y

E. Q6

Question 7

```
class M {
    int x;
    static int y = 1;

    M(int x) {
        this.x = x;
        ++y;
    }
}

class StaticQ12 {
    public static void main(String[] args) {
        M m1 = new M(10);
        M m2 = new M(20);
        m2.y++;
        m1.x++;

        System.out.println(m1.x + " " + m2.x + " " + m1.y + " " + m2.y);
    }
}
```

What is the output of the following segment of code?

A. 11 20 4 4

B. 11 20 3 3

C. 11 20 2 3

D. 11 21 2 3

E. 11 11 3 3

Question 8

Which of the following statements is true?

A. The super keyword can be used to call a method in any other class of the program

B. The super keyword can be used to speed up the execution of the current class.

C. The super keyword can be used to call a constructor of the parent class, to call a method of the parent class or to access a field of the parent class.

D. The super keyword can be used to give access to "hidden" parts of other classes

E. The super keyword can be used to call a method of the parent class or to access a field of the parent class. It cannot be used to call a constructor of the parent class.

Question 9

```
class RefsQ11 {
    public static void main(String[] a) {
        String x = new String("a dog ");
        String y = "a cat ";
        String z = x + y;
```

```
        x = z;
        x = y;

        System.out.println(z + "having a nice day");
    }
}
```

What is the output of the following segment of code?

A. a dog having a nice day

B. a dog a cat having a nice day

C. a cat having a nice day

D. a dog a dog having a nice day

E. a cat a dog having a nice day

Question 10

```
class MyNumber {
    int i = 5;
    MyNumber(int i) {
        this.i = i;
    }
}

class Q5 {
    void modify(int i) {
        i++;
    }
    void modify(MyNumber m) {
        m.i++;
    }
    public static void main(String[] args) {
        int i = 4;
        Q5 q5 = new Q5();
        q5.modify(i);
        MyNumber m = new MyNumber(2);
        q5.modify(m);
        System.out.println(i + " " + m.i);
    }
}
```

What is the output of the following segment of code?

A. 3 3

B. 4 3

C. 4 5

D. 3 4

E. 2 4

Question 11

```
class Parent {
    Parent() {
        System.out.print("Parent ");
    }
}

class Child extends Parent {
    Child() {
        System.out.print("Child ");
```

```
    }
}

class GrandChild extends Child {
    GrandChild() {
        System.out.print("Grandchild ");
    }

    GrandChild(int x) {
    }
}

class Q14 {
    public static void main(String[] args) {
        Parent p = new GrandChild();
        Parent p2 = new GrandChild(5);
        System.out.println();
    }
}
```

What is the output of the following segment of code?

- **A.** Grandchild Child Parent Child Parent
- **B.** Parent Child Grandchild Parent Child Grandchild
- **C.** *(remaining options not visible in the screenshot)*

Question 12 (number not visible)

```
    public static void main(String[] args) {
        A a1 = new A();
        a1.i = 6;   // line 6
        a1.j = 7;   // line 7
        a1.k = 8;   // line 8
    }
}

File A.java contains:

package f1;

public class A {
    int i;
    protected int j;
    private int k;
}

File B.java contains:

package p2;

import f1.A;

class B extends A {
    void foo() {
        i = 5;    // line 15
        j = 10;   // line 16
        k = 20;   // line 17
    }
}
```

Which lines of the following code will cause a compilation error? (The first lines of the question,

including the file/package of the class containing main, were cut off in the screenshot.)

A. lines 8, 15, 16, 17

B. lines 8, 16 and 17

C. lines 8, 15 and 17

D. lines 8 and 17

E. lines 15 and 17

Question 13

```
class D {
    private int x;
    private int y = 5;

    public static void main(String[] args) {
        D a1 = new D();
        D a2 = new D();
        D a3 = new D();
        a1.x = a3.y;
        a3 = a1;
        a1 = a3;
        a2.x = 12;
        a3.x = a3.x + 2;
        a1.x = 30;
        a1.y = 21;
        a2.y = 22;
        System.out.println(a1.x + " " + a2.x + " " + a3.x + " " + a1.y +
                           " " + a2.y);
    }
}
```

What is the output of the following code?

A. 30 30 22 22 21

B. 30 30 30 21 22

C. 30 30 1 22 22

D. 30 12 30 21 22

E. 22 30 1 11 22

Question 15

```
class References {
    int x = 8;
    int y = 1;

    public static void main(String[] args) {
        References r1 = new References();
        r1.y = 10;
        References r2 = new References();
        r2.y = ++r1.x;

        r1 = r2;
        ++r2.y;

        References r3 = new References();
        r2 = r3;
        System.out.println(r1.x + " " + r1.y + " " + r2.x + " " + r2.y);
    }
}
```

What is the output of the following code?

A. 8 10 8 1

B. 8 1 8 1

C. 9 10 8 1

D. 10 8 8 10

Question 16

```
class A2 {
    public static void main(String args[]) {
        String string = "STRING";
        if ("STRING".equals("STRING"))
            System.out.print("Equal ");
        else
            System.out.print("Not Equal ");

        if (string == "STRING")
            System.out.print("Equal ");
        else
            System.out.print("Not Equal ");
    }
}
```

What is the output of the following segment of code?

A. Not Equal Equal

B. Not Equal Not Equal

C. Equal Not Equal

D. Equal Equal

Part B - Programming (40%)

You are expected to use a pen and paper, write the source code, scan or photograph it with a mobile phone and create a single PDF file (named with your UoW ID, e.g. w1678756.pdf) with correct orientation and clarity. You are not allowed to use the keyboard and type in any program.

Implement the Java classes (with full details of all fields and methods, including constructors and getter and setter methods) which are required to simulate the following problem:

A casino has 10 Fruit Machines that customers are using to play. A fruit machine is a machine in which a player (customer) inserts a coin (worth 1 pound) every time to play a single game. Following the insertion of a coin, the machine displays to the customer exactly 3 random fruits next to each other (each fruit is chosen randomly from a total of 21 fruits; see below for the probabilities). If during a single game the machine displays 2 fruits which are the same, the player is credited with 20 pounds. If during a single game the machine displays all 3 fruits the same, the player is credited with 100 pounds in their account.

The customer (player) has a name, an address, lives in a city (which might be different from that of the casino), has a casino balance (credit), which is the money available to play, and has a favourite casino with a specific name. A casino has a location (city) and can accommodate up to 100 customers at the same time.

A fruit machine holds a number of coins (1 coin is worth 1 pound). Each fruit machine displays 2 fruits of the same kind with probability 10% if the credit in the machine (number of coins) is greater than 300 (if the credit is less than that, it will never display 2 fruits which are the same). A fruit machine displays 3 fruits of the same kind with probability 2% if the credit in the machine is greater than 500 (if the credit is less than that, it will never display 3 fruits which are the same). If the credit in the machine is less than 300, the fruit machine always displays 3 unique fruits.

As part of your implementation, in addition to the classes described above, you should also provide a Simulation class in which a player interacts with a fruit machine and plays 100 games with it. You do not need to include getters and setters for every single field: assuming you implement the constructor of a class to initialise the fields, you could just provide getters and setters for 1 or 2 fields only.

Answer Key (confirmed by running the code)

- **Q1: D** - calling run() directly runs T1's infinite loop on the main thread, so t2.start() is never reached; t1.start() starts it as a separate thread.
- **Q2: A** - instructional decomposition (the OOP characteristics are polymorphism, abstraction, encapsulation and inheritance).
- **Q5: B** - false (Object.equals compares references) then true (String.equals compares contents).
- **Q6: B** - Q6 Y Q6 (dynamic dispatch uses the object's real type).
- **Q7: A** - 11 20 4 4 (y is static and shared: 1 → 2 → 3 → 4).
- **Q8: C**
- **Q9: B** - a dog a cat having a nice day (reassigning x does not change z).
- **Q10: B** - 4 3 (the int is passed by value; the object's field is changed).
- **Q11:** Parent Child Grandchild Parent Child (GrandChild(int) still implicitly calls super(); the matching option was not visible).
- **Q12: C** - lines 8, 15 and 17 (private k is never accessible; package-private i is not visible from package p2; protected j is fine in the subclass). Confirmed with the class containing main in package f1, which the answer options imply.
- **Q13: D** - 30 12 30 21 22 (a1 and a3 refer to the same object).
- **Q15: A** - 8 10 8 1
- **Q16: D** - Equal Equal (both literals are the same interned String, so == is also true).